

Translation Quality and Named-Entity Preservation in Ukrainian–English Machine Translation

Katsutoshi Fujita, Anastasiia Neskoro dieva, Joel Vazquez, Kajsa Vesterberg
Uppsala University

{katsutoshi.fujita.0614, anastasiia.neskorodieva.3784, joel.vazquez.3528, kajsa.vesterberg.0919}@student.uu.se

Abstract

This paper investigates the performance of different machine translation systems for Ukrainian–English named entity translation. The models were trained using a Ukrainian–English parallel dataset with a BPE size of 30,000 and evaluated both quantitatively using BLEU and SacreBLEU and through qualitative analysis. The OPUS-MT model performed the best quantitatively, receiving a SacreBLEU score of 28.37, whereas the NLLB-200 model performed the best qualitatively, with the most correctly translated named entities. In conclusion, large-scale pretrained models tend to outperform smaller trained models, with NLLB-200 performing the best overall.

1 Introduction

Machine translation (MT) is often evaluated with automatic metrics such as BLEU, which measures the similarity between a system output and a reference translation. While such metrics are useful for estimating overall translation quality, they do not always show whether a system preserves key information. Furthermore, this can extend to affect acronyms, abbreviations, and geographical names. In Ukrainian–English translation, items such as ‘OON’ (UN), ‘VIL’ (HIV), ‘ZSU’ (Armed Forces of Ukraine), and ‘Kyiv’ can carry important information in a sentence; if such items are omitted or mistranslated, the output can lead to being factually misleading even if parts of the translation remain fluent.

This issue is particularly important in low-resource machine translation tasks, as neural machine translation (NMT) systems trained from scratch often require large amounts of parallel data in order to induce reliable patterns. When only limited data is available, the

result can be that the systems produce repetitive output, therefore failing to preserve rare or highly specific terms, in this case, proper nouns.

In this project we investigate Ukrainian-to-English machine translation with a focus on acronym and named-entity preservation, examining the issue from two perspectives. First we study the effect of subword preprocessing by training a Fairseq transformer model with a different vocabulary size of BPE. Second, we compare different MT system types: a Moses statistical machine translation (SMT), the best-performing Fairseq transformer model, and two pretrained MT systems. This allows a comparison of a traditional phrase-based system, a neural system trained from scratch on the provided data, and models that benefit from larger multilingual pretraining. In addition to BLEU, we perform targeted manual analysis of selected examples containing acronyms and named entities.

Our study is guided by the following research questions:

1. How do preprocessing and model type affect the translation of Ukrainian acronyms and location expressions?
2. How accurately do Ukrainian–English MT systems preserve and translate proper nouns in news translation?
3. To what extent does BLEU reflect factual preservation errors involving acronyms and named entities?

2 Background

Ukrainian is an East Slavic language written in Cyrillic; it has richer inflectional morphology than English, including grammatical case,

number, and gender. This rich morphology affects common noun forms and can thus also be found in proper and geographic names. As a result, the same named entity may appear in different surface forms depending on the grammatical role in the sentence. This is relevant for Ukrainian–English MT because the system needs to recognize the different Ukrainian grammatical forms that can refer to the same entity and render them appropriately in English.

Machine translation systems are commonly evaluated using BLEU, an automatic metric based on n-gram overlap between system output and reference translation (Papineni et al., 2002). BLEU is useful for comparing overall system performance, but it does not directly evaluate whether specific factual items, such as names, acronyms, or organizations, are preserved correctly. For this reason, targeted analysis is useful when studying errors involving named entities and acronyms.

Subword segmentation is widely used in neural machine translation to reduce vocabulary sparsity. Sennrich et al. (2016) introduced the use of Byte Pair Encoding (BPE) for NMT to handle rare and unknown words by representing them as sequences of subword units. This is especially relevant for morphologically rich languages, where many word forms may occur only rarely in the training data. However, BPE can also split acronyms and named entities. A smaller BPE vocabulary produces more segmentation, while a larger vocabulary preserves more whole tokens. This creates a trade-off between generalization and entity preservation.

In our experiments, we compare statistical and neural MT systems. Moses is a widely used phrase-based statistical MT toolkit (Koehn et al., 2007) and can be competitive in limited data settings because it relies on phrase correspondences learned from parallel data. In contrast, our neural MT models are Transformer models trained with Fairseq. The Transformer architecture was introduced by Vaswani et al. (2023), and Fairseq is a sequence modeling toolkit commonly used for training translation models. Since Transformers trained from scratch generally require substantial parallel data, they may struggle in limited resource settings.

We also compare these models with pre-trained multilingual MT systems, such as

NLLB and OPUS-MT. NLLB was designed to support translation for many languages through large-scale multilingual training and transfer (Team et al., 2022), whereas OPUS-MT provides open translation models trained on OPUS parallel corpora (Tiedemann and Thottingal, 2020). These pretrained systems provide a useful contrast with models trained only on the project dataset.

In Konyk et al. (2023), the authors examine how a sequence-to-sequence LSTM performs in Ukrainian-to-English translation compared to large established tools like Google Translate. They suggest that targeted deep learning can outperform these tools, as it utilizes sentence-wide context instead of word-to-word translations. The authors highlighted how even smaller, less complex models can be useful for everyday translations, thus motivating the use of simpler models for this paper.

As in all translations, it is important to remember the challenges faced when translating from one language to another. Styrnik (2023) highlights some Ukrainian-to-English specific issues, such as the free word order of Ukrainian, the different case systems, and the few verb forms found in Ukrainian having to become more complex when translated into English. Correspondingly, Kushnirova et al. (2025) addresses similar issues but focuses on lexical transformations and contextual substitutions. These concepts are the techniques used to identify the meaning of a word in context, rather than using one-to-one translations. The author(s) state that some of these core transformations are generalization, semantic development, compensation of losses, and addition and omission of words. Though these concepts are focused around English-to-Ukrainian translations in the paper, the issues with translation are equally relevant in Ukrainian-to-English translation, particularly in the preservation of named entities.

3 Method

3.1 Data and Preprocessing

We used the TED Talks multilingual corpus dataset originally provided for the 2025 WMT conference.¹ The dataset consists of presplit

¹<https://www2.statmt.org/wmt25/limited-resources-slavic-llm.html>

English and Ukrainian text data for training, development (dev), and testing. The number of sentences in each split is as follows: train: 108,496; dev: 3,061; test: 3,752.

The dataset consists purely of raw data with no tokenization or BPE preprocessing applied. Since we planned to conduct experiments with several models, such as an SMT model, an NMT model with BPE-segmented data, and multilingual pretrained models. We preprocessed the data to create both tokenized data and BPE-segmented data for NMT. Though it is possible to do so, the SMT pipeline did not utilize BPE preprocessing, as Moses is phrase-based and operates at the word level. BPE preprocessing would instead create much larger and sparser phrase tables, and its reordering model relies on word boundaries, which would be disrupted by subword splitting, thus yielding a much worse result compared to using only tokenized data. As the tokenizer, we used the tokenizer from mosesdecoder.² The restrictions of the tokenizer are as follows: sentences containing fewer than one word or more than 175 words are excluded; if the length ratio between the Ukrainian and English sentences differs by a factor of three or more, the pair is excluded.

For BPE segmentation, in addition to the tokenization process described above, we further preprocessed the data using Python code from subword-nmt.³ Setting the vocabulary size to 30,000, we created BPE-segmented data by training BPE algorithms with subword segmentation rules common to both languages (the top 30,000 most frequently occurring character combinations). In addition to the Fairseq Transformer models, we also trained a simple sequence-to-sequence baseline using the BPE 30k data. This model was included as a lower-bound neural baseline, allowing us to compare the Transformer architecture with a simpler neural MT system trained on the same subword-preprocessed corpus.

For the pretrained systems, we used OPUS-MT / Helsinki-NLP and NLLB-200. For NLLB-200, we used the HuggingFace model facebook/nllb-200-distilled-600M in a zero-shot setting, without fine-tuning it on our project dataset. This allowed us to compare

models trained only on the project data with pretrained multilingual systems.

3.2 BPE Vocabulary Size Experiments

This experiment explores how BPE vocabulary size affects Ukrainian–English Fairseq Transformer models, especially for acronyms and named entities such as ‘США,’ ‘ООН,’ ‘MIT,’ ‘Нью-Йорк,’ and ‘Швеція’. We trained five Transformer models from scratch using the same train, validation, and test splits but with different BPE sizes: 5k, 10k, 15k, 30k, and 45k. We trained the main Transformer models from scratch with BPE sizes of 5k, 10k, and 30k using the same train, validation, and test splits, and additionally ran exploratory 5-epoch experiments with 15k and 45k BPE sizes.

All models were trained in Fairseq using the transformer_iwslt_de_en architecture, Adam optimization, label-smoothed cross entropy, dropout, and BLEU-based checkpoint selection. The only main difference between the systems was the BPE vocabulary size.

Smaller BPE vocabularies split words into more subword units, which can aid generalization but may fragment names and acronyms. Larger BPE vocabularies preserve more complete word forms, which may help factual items remain intact but can also increase sparsity.

To analyze this, we checked 206 selected acronym and named-entity occurrences from the test set and counted whether each item was kept whole or split across the different BPE settings:

BPE size	Whole	Split	Total	Whole %
5k	28	178	206	13.6%
10k	100	106	206	48.5%
30k	160	46	206	77.7%

Table 1: Named-entity segmentation by BPE vocabulary size.

The results show a clear effect of BPE size. The 5k model split most selected items, preserving only 13.6% as whole units, while the 30k setting preserved 77.7% of them. This suggests that larger BPE vocabularies are more suitable when the analysis focuses on preserving acronyms and named entities, although this should be interpreted as a targeted segmentation analysis rather than a complete named entity evaluation.

²<https://github.com/moses-smt/mosesdecoder.git>

³<https://github.com/rsennrich/subword-nmt.git>

Model	BLEU-4	SacreBLEU
SMT	26.20	26.3
NMT (40 epochs)		
BPE 5k	8.00	6.3
BPE 10k	8.45	6.9
BPE 30k	29.66	26.7
NMT (5 epochs)		
BPE 5k	6.33	–
BPE 10k	8.04	–
BPE 15k	5.08	–
BPE 30k	8.58	–
BPE 45k	5.90	–
OPUS-MT	28.37	28.37
NLLB-200	28.19	28.19
Seq2Seq	7.17	7.2

Table 2: BLEU-4 and SacreBLEU scores for each model.

4 Results

The BLEU results showed (Table 2) a clear advantage for larger BPE vocabularies. The 30k BPE model substantially outperformed the smaller models, reaching 29.66 test BLEU, compared with 8.00 for BPE 5k and 8.45 for BPE 10k. This suggests that the smaller BPE vocabularies fragmented the data too aggressively, limiting lexical preservation and translation quality. The 5k model also showed signs of over-generation, with a much higher output-to-reference-length ratio. These results are consistent with the acronym and named-entity analysis: BPE 30k preserved 77.7% of selected items as whole units, while BPE 5k preserved only 13.6%. The Seq2Seq baseline achieved substantially lower performance, with 7.17 BLEU and 7.2 sacreBLEU. This indicates that BPE pre-processing alone was not sufficient for strong NMT performance.

4.1 Annotation Results

We conducted a human annotation process by randomly selecting 50 translated sentences and having two Ukrainian speakers (one native and one non-native) assess whether proper nouns were translated appropriately. We introduced the following annotation guideline: we annotate only the target item (acronym, proper noun, etc.), not overall sentence quality, using the labels CORRECT (all or almost all target items correct), PARTIAL (at least half of target items correct or recognizable), INCORRECT (most target items wrong, distorted, or omitted), and OMITTED (all target items omitted).

We annotated generated translations from

Moses SMT, the NMT Fairseq model, OPUS-MT/Helsinki-NLP, and NLLB-200. We calculated Cohen’s kappa as an inter-annotator agreement score.

Model	Cohen’s κ
Moses SMT	0.88
NMT Fairseq	0.87
OPUS-MT	0.93
NLLB-200	0.92

Table 3: Inter-annotator agreement (Cohen’s κ) per model.

According to Landis and Koch (1977), each score corresponds to “Almost Perfect” agreement, indicating that the annotation guideline was effective and the results are reliable.

Model	C	P	I	O
Moses SMT	27	8	11	0
NMT Fairseq	23	11	10	1
OPUS-MT	33	10	4	1
NLLB-200	43	6	0	0

Table 4: Agreed annotation label counts per model. C = CORRECT, P = PARTIAL, I = INCORRECT, O = OMITTED.

5 Discussion

The results suggest that pretrained multilingual models are more reliable for preserving acronyms and named entities than systems trained only on the project data. NLLB-200 performed best, with 43 agreed CORRECT labels and no agreed INCORRECT or OMITTED cases, followed by OPUS-MT. Moses SMT and the Fairseq Transformer produced more incorrect target-item translations.

This is likely because NLLB-200 and OPUS-MT benefit from large-scale multilingual pre-training and have been exposed to a wider range of names, acronyms, organizations, and geographical expressions. In contrast, Moses SMT and Fairseq relied only on the provided project data, making rare or specific target items harder to translate reliably.

The annotation also suggests that shorter and internationally common acronyms were generally easier for the models to preserve. Items such as ‘UN’, ‘EU’, ‘TED’, ‘SQL’, ‘MRI’, and ‘HIV’/‘AIDS’ were often translated or

copied correctly. In contrast, longer proper names, organization names, and mixed examples containing several target items were more error-prone, often involving spelling distortions, partial translation, or omission of one or more entities.

These findings also illustrate why BLEU alone is not sufficient for this task. BLEU and sacreBLEU measure n-gram overlap between the model output and the reference translation, capturing general similarity but not whether a specific acronym or named entity was preserved correctly. Manual annotation is therefore necessary to evaluate target items directly.

The BPE experiments show that preprocessing also matters. Larger BPE vocabularies preserved more selected target items as whole units and improved the Fairseq model's BLEU score. However, even the best Fairseq model remained weaker than the pretrained systems in the manual annotation. This suggests that BPE size can improve entity preservation, but large-scale pretraining plays a stronger role in this task.

Limitations

Firstly, it needs to be mentioned that this is a small study heavily dictated by a time constraint; it is still important to point out its issues, the main one being the size of the manually annotated sample. Due to only two authors of this paper being speakers of Ukrainian and also having five model outputs to annotate, only a small portion of the produced data could be evaluated. This could be addressed by having more annotators working on the datasets and also having a larger portion of data be annotated. Because of these constraints, the qualitative analysis results should be interpreted as a targeted analysis rather than a full evaluation of all named entities in the dataset. Furthermore, the dataset picked was adequate for the task at hand, but not optimal for a larger study. Therefore, using a different, more varied dataset with more target entities would be advised for future work.

Additionally, computational constraints caused by the cluster instability on UPPMAX limited the number of model configurations that could be evaluated, preventing all BPE sizes from being trained for 40 epochs. While

the overall conclusions remain consistent with what a more extensive evaluation would likely yield, the results could have been more extensive under more stable cluster conditions.

Conclusion

Overall, the results suggest that pretrained multilingual models, especially NLLB-200, are more reliable for preserving acronyms and named entities in Ukrainian–English translation. While BLEU and sacreBLEU are useful for measuring general translation quality, the manual annotation more directly revealed whether specific target items were translated or preserved correctly.

For future work, the analysis could be extended to a larger and more acronym-dense dataset. It would also be beneficial to annotate more examples and compare results by target-item type, such as acronyms, personal names, organizations, and geographical names. Future studies could furthermore examine whether fine-tuning pretrained models or using named-entity-aware preprocessing improves the translation of rare and domain-specific entities.

References

- Philipp Koehn, Hieu Hoang, Alexandra Birch, Chris Callison-Burch, Marcello Federico, Nicola Bertoldi, Brooke Cowan, Wade Shen, Christine Moran, Richard Zens, Chris Dyer, Ondřej Bojar, Alexandra Constantin, and Evan Herbst. 2007. [Moses: Open source toolkit for statistical machine translation](#). In Proceedings of the 45th Annual Meeting of the Association for Computational Linguistics Companion Volume Proceedings of the Demo and Poster Sessions, pages 177–180, Prague, Czech Republic. Association for Computational Linguistics.
- Myroslav Konyk, Victoria Vysotska, Svitlana Goloshchuk, Roman Holoshchuk, Sofia Chyrun, and Ihor Budz. 2023. [Technology of ukrainian-english machine translation based on recursive neural network as lstm](#). In International Conference on Computational Linguistics and Intelligent Systems.
- T. Kushnirova, Yu Nikolaienko, and M. Bochkar. 2025. [Lexical and grammatical features of translation of english journalistic texts into ukrainian in the process of intercultural communication](#). International Humanitarian University Herald. Philology, pages 240–243.
- J Richard Landis and Gary G. Koch. 1977. [The](#)

measurement of observer agreement for categorical data. *Biometrics*, 33 1:159–74.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. [Bleu: a method for automatic evaluation of machine translation](#). In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA. Association for Computational Linguistics.

Rico Sennrich, Barry Haddow, and Alexandra Birch. 2016. [Neural machine translation of rare words with subword units](#). In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1715–1725, Berlin, Germany. Association for Computational Linguistics.

N. S. Styrnik. 2023. [Innovations in english grammar and vocabulary: The challenges of translation into ukrainian](#). In *Неофілологічні студії*, pages 150–153. Dnipro.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, Kenneth Heafield, Kevin Heffernan, Elahe Kalbassi, Janice Lam, Daniel Licht, Jean Maillard, Anna Sun, Skyler Wang, Guillaume Wenzek, Al Youngblood, Bapi Akula, Loic Barrault, Gabriel Mejia Gonzalez, Prangthip Hansanti, and 20 others. 2022. [No language left behind: Scaling human-centered machine translation](#). Preprint, arXiv:2207.04672.

Jörg Tiedemann and Santhosh Thottingal. 2020. [OPUS-MT – building open translation services for the world](#). In *Proceedings of the 22nd Annual Conference of the European Association for Machine Translation*, pages 479–480, Lisboa, Portugal. European Association for Machine Translation.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2023. [Attention is all you need](#). Preprint, arXiv:1706.03762.